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Important Formulas: The BAUDHĀYANA- Pythagoras Theorem

Table of Contents

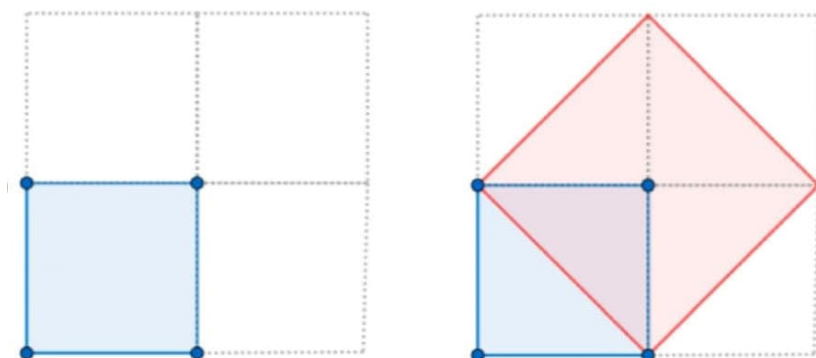
1. Doubling and Halving Squares
2. Isosceles Right Triangle
3. The Baudhāyana-Pythagoras Theorem
4. Special Value: $\sqrt{2}$
5. Baudhāyana Triples

[View More](#) 

Important Formulas: The BAUDHĀYANA- Pythagoras Theorem

Doubling and Halving Squares

Doubling a Square



Formula: If a square has side length s , a square constructed on its diagonal has area equal to **double** the original area.

Steps:

1. Original square area = s^2
2. Diagonal of square = d
3. New square area = $d^2 = 2s^2$

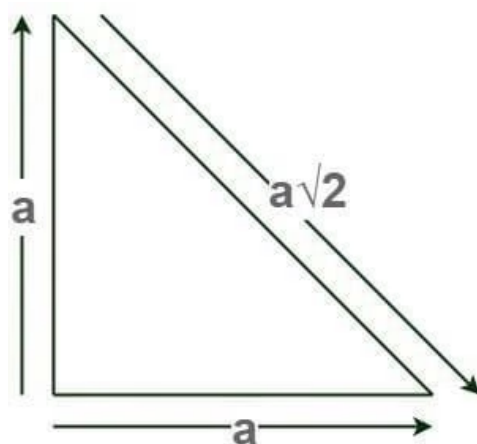
Key Relation: $d^2 = 2s^2$

Halving a Square

Formula: To create a square with half the area, construct a tilted square inside using midpoints.

Relation: If original area = A , then new area = $A/2$

Isosceles Right Triangle



Hypotenuse Formula

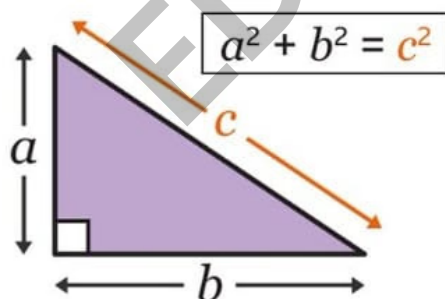
Let a = length of equal sides

Let c = length of hypotenuse

Formula: $c^2 = 2a^2$

Therefore: $c = a\sqrt{2}$

The Baudhāyana-Pythagoras Theorem



Main Theorem

For a right-angled triangle with:

Shorter sides of length **a** and **b**

Hypotenuse of length **c**

Formula: $a^2 + b^2 = c^2$

Finding Missing Sides:

When hypotenuse is unknown:

$$c = \sqrt{a^2 + b^2}$$

When one shorter side is unknown:

$$a = \sqrt{c^2 - b^2}$$

$$b = \sqrt{c^2 - a^2}$$

Verification Example:

If $a = 3$ and $b = 4$:

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$c = \sqrt{25} = 5$$

Special Value: $\sqrt{2}$

Decimal Approximation

$$\sqrt{2} \approx 1.41421356...$$

Bounds on $\sqrt{2}$:

Rough bounds: $1 < \sqrt{2} < 2$

Closer bounds: $1.4 < \sqrt{2} < 1.5$

Closer still: $1.41 < \sqrt{2} < 1.42$

Very close: $1.414 < \sqrt{2} < 1.415$

Key Property:

$\sqrt{2}$ **cannot** be expressed as a terminating decimal

$\sqrt{2}$ **cannot** be expressed as a fraction m/n (where m, n are counting numbers)

$\sqrt{2}$ has a non-terminating, non-repeating decimal representation

Baudhāyana Triples

Definition

A Baudhāyana triple (a, b, c) consists of three positive integers where:

$$a^2 + b^2 = c^2$$

Common Baudhāyana Triples:

$(3, 4, 5)$

$(5, 12, 13)$

$(8, 15, 17)$

$(7, 24, 25)$

$(12, 35, 37)$

$(15, 36, 39)$

Primitive vs Non-Primitive Triples

Primitive Triple: No common factor greater than 1

Example: $(3, 4, 5)$ is primitive

Non-Primitive Triple: Has a common factor > 1

Example: $(6, 8, 10) = 2 \times (3, 4, 5)$ is non-primitive

Scaling Formula

If (a, b, c) is a Baudhāyana triple, then (ka, kb, kc) is also a Baudhāyana triple for any positive integer k .

Proof:

$$(ka)^2 + (kb)^2 = k^2a^2 + k^2b^2 = k^2(a^2 + b^2) = k^2c^2 = (kc)^2$$

Examples of Scaled Triples:

From (3, 4, 5):

$$k = 2: (6, 8, 10)$$

$$k = 3: (9, 12, 15)$$

$$k = 4: (12, 16, 20)$$

$$k = 10: (30, 40, 50)$$

$$k = 100: (300, 400, 500)$$

Generating Primitive Triples**Using Consecutive Odd Numbers**

Formula: $(n - 1)^2 + (2n - 1) = n^2$

Where: $2n - 1$ is the n th odd number

Method:

If $2n - 1$ is a perfect square, we get a Baudhāyana triple.

Examples:

1. **9 is the 5th odd number** ($9 = 2 \times 5 - 1$)

$$n = 5$$

$$(5 - 1)^2 + 9 = 5^2$$

$$4^2 + 3^2 = 5^2 \rightarrow \text{Triple: } (3, 4, 5)$$

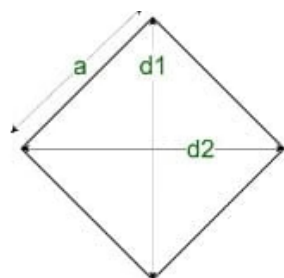
2. **25 is the 13th odd number** ($25 = 2 \times 13 - 1$)

$$n = 13$$

$$(13 - 1)^2 + 25 = 13^2$$

$$12^2 + 5^2 = 13^2 \rightarrow \text{Triple: } (5, 12, 13)$$

Special Applications**Rhombus Diagonal Formula****Area of Rhombus**



Rhombus

$$\text{Area of Rhombus} = \frac{(d_1 * d_2)}{2}$$

$$\text{Diagonal } d_2 = \sqrt{4a^2 - d_1^2}$$

$$\text{Diagonal } d_1 = \sqrt{4a^2 - d_2^2}$$

For a rhombus with diagonals d_1 and d_2 :

The diagonals bisect each other at right angles

$$\text{Each side } s = \sqrt{[(d_1/2)^2 + (d_2/2)^2]}$$

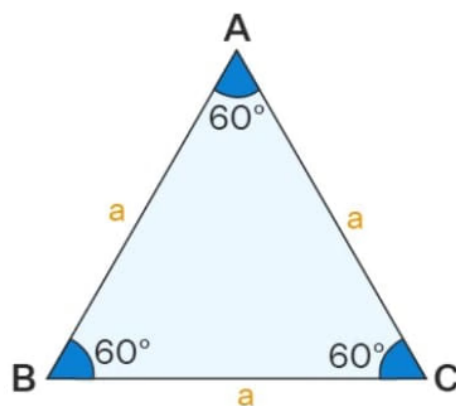
Example: If diagonals are 24 and 70:

$$s^2 = 12^2 + 35^2 = 144 + 1225 = 1369$$

$$s = 37 \text{ units}$$

Equilateral Triangle Area

Area of Equilateral Triangle



$$\text{Area of Equilateral Triangle} = \frac{\sqrt{3}}{4} a^2$$

For an equilateral triangle with side s :

$$\text{Height } h = (s\sqrt{3})/2$$

$$\text{Area} = (s^2\sqrt{3})/4$$

Example: If $s = 6$:

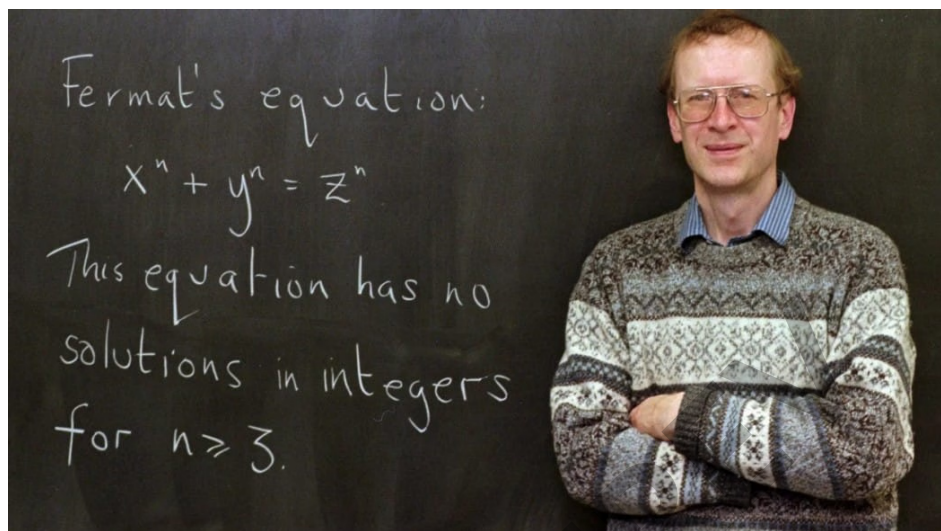
$$h^2 + 3^2 = 6^2$$

$$h^2 = 36 - 9 = 27$$

$$h = \sqrt{27} = 3\sqrt{3}$$

$$\text{Area} = (6 \times 3\sqrt{3})/2 = 9\sqrt{3} \text{ square units}$$

Fermat's Last Theorem



Statement

The equation $x^n + y^n = z^n$ has **NO solution** in positive integers when $n > 2$.

For $n = 2$: Infinite solutions (Baudhāyana triples)

For $n = 3, 4, 5, \dots$: NO solutions exist

Proven by: Andrew Wiles in 1994 (after 300+ years of attempts)

